

10/11/2026

TryHackMe - Critical

Read the intro first. U'll know,
The memory dump file is memdump.mem already sitting on the lab machine at /home/analyst.
Volatility 3 is set up as an alias - just type vol.

Task 2 - Memory Forensics:

Just theory. Memory forensics = analyzing RAM. RAM is volatile so it gets wiped on shutdown.
Two phases - Memory Acquisition (grab the dump) and Memory Analysis (study it). In our scenario the dump is already acquired.
Answers - volatile memory / RAM, and Memory Acquisition.
(in detailed in CTF101 writeup)

Task 3 - Environment & Setup:

Imaging tools by OS :-
Windows -> FTK Imager, WinPmem
Linux -> LIME
macOS -> osxpmem
Volatility help menu :-
 vol -h
List Windows plugins :-
 vol windows --help
Answers - windows.info, LIME, vol -h.

Task 4 - Gathering Target Information:

Use windows.info plugin to get the OS info, architecture, kernel base etc.
 vol -f memdump.mem windows.info
From the output check :-
Is64Bit - says True or False. If True the machine is x64.
PE MajorOperatingSystemVersion / Minor - Windows version like 10.0
Kernel Base - hex address starting with 0x
Note - use ur OWN dump's output values by using given lab machine, not the example in the room docs.

Task 5 - Searching for Suspicious Activity:

Check network connections first :-
 vol -f memdump.mem windows.netscan
Look for foreign IPs on suspicious ports like 3389 (RDP) or 80 (HTTP).
We found RDP from 192.168.182.139 (attacker's initial access) and a port 80 connection to 192.168.182.128 owned by updater.exe.

Then check the process tree :-

```
vol -f memdump.mem windows.pstree
```

Look for process names that don't belong to Windows.

We found critical_updat (truncated, real name critical_update.exe) and its child updater.exe with PID 1612. Both not Windows processes - so malicious.

Note - pstree truncates names to 14 characters. critical_update.exe shows as critical_updat.

Task 6 - Finding Interesting Data:

Now go deep on updater.exe (PID 1612).

Find where it lives on disk :-

```
vol -f memdump.mem windows.filescan > filescan_out
(why?, cause the output will be very large hard to read put all the data in
the file and then use grep)
cat filescan_out | grep critical
cat filescan_out | grep updater
```

Path found - C:\Users\user01\Documents\updater.exe

Get detailed file timestamps (MFT):-

```
vol -f memdump.mem windows.mftscan.MFTScan > mftscan_out
cat mftscan_out | grep important
```

Dump the process's memory :-

```
vol -f memdump.mem -o . windows.memmap --dump --pid 1612
```

Output is a file called pid.1612.dmp

Extract readable text from the dump :-

```
strings pid.1612.dmp | grep critical-update
strings pid.1612.dmp | grep -i "server:"
strings pid.1612.dmp | grep -B 10 -A 10 "encKEY.txt"
```

What we found :-

Attacker URL - <http://key.critical-update.com/encKEY.txt>

Encryption key - cafebabe

HTTP Server header - SimpleHTTP/0.6 Python/3.10.4 (attacker used Python's built-in http.server module)

Encrypted file - important_document.pdf

'SimpleHTTP/0.6 Python/3.10.' is the ans

done!!